Notes: Zheng (JF, 1999)

Is Money Smart? A Study of Mutual Fund Investors' Fund Selection Ability

Contents

1 Big picture

Zheng (1999) asks whether mutual fund investors are able to select funds that will subsequently perform well. The paper studies this question from the perspective of **investor cash flows**, not from the perspective of fund managers alone. If new money flows into funds that later outperform funds losing money, then aggregate investors display a "smart money" effect.

The central finding is nuanced. Funds receiving more new money subsequently outperform funds losing money, so there is evidence of fund selection ability in relative terms. However, for the full sample, there is little evidence that simply investing in funds with positive new money beats the market after standard risk adjustment. The stronger, more practically useful evidence is concentrated in **small funds**, especially small no-load funds, and the effect is short-lived.

The paper is therefore not a broad claim that mutual fund investors can beat the market. It is a more precise claim: aggregate fund flows contain short-horizon information about future relative fund performance, and this information is strongest among small funds.

2 Incremental contribution of the paper

1. **Moves from manager skill to investor selection skill.** Much of the mutual fund literature asks whether fund managers beat benchmarks. Zheng asks whether investors identify better funds when allocating new money.
2. **Separates the smart money effect from the information effect.** The smart money effect only requires that new money flows predict future relative performance. The information effect additionally requires that a feasible trading strategy based on flows earns abnormal returns.
3. **Uses a broad survivorship-aware mutual fund sample.** The sample covers diversified U.S. equity funds from January 1970 to December 1993, includes load and no-load funds, and includes defunct, merged, and newly created funds.
4. **Uses the Grinblatt–Titman benchmark-free measure for aggregate investor reallocations.** This test asks whether changes in investors' aggregate portfolio weights predict future fund returns without relying on a specific benchmark model.
5. **Evaluates implementable and non-implementable strategies separately.** The paper compares GT zero-cost portfolio tests, long-short positive-minus-negative new-money portfolios, and long-only positive-flow portfolios.
6. **Controls for style and macro conditioning.** Conditional one-factor and Fama–French three-factor alphas show that the effect is not explained away by public macro variables or time-varying style exposure.
7. **Identifies where the effect lives.** The smart money effect is mainly in small funds; it is weaker for large funds, stronger for load funds before loads are deducted, and still partly implementable for small no-load funds.
8. **Shows that the effect is short-lived.** The predictive content of flows fades with the holding period and reverses by roughly 30 months.

3 Main research question

The paper's core question is:

> Can mutual fund investors forecast future fund performance through their buying and selling decisions?

The question matters because open-end mutual funds were already a major investment vehicle by the 1990s. If investors systematically reallocate toward future winners, their realized performance can be better than the performance of the average fund. This helps explain why actively managed funds can keep receiving assets even if the average fund does not beat passive benchmarks.

4 Key concepts

4.1 Smart money effect

The smart money effect means that new money flows are positively related to subsequent fund performance. A minimal test is whether funds with positive new money outperform funds with negative new money in the next quarter. This is a relative performance concept and does not require the positive-flow funds to beat the market.

4.2 Information effect

The information effect is stronger. It asks whether public information about new money flows can be used to earn abnormal returns. This requires an implementable strategy, allowing for reporting delays and trading restrictions. A long-short portfolio that shorts mutual funds is generally not implementable, so the paper is careful not to treat every positive long-short result as a practical trading opportunity.

4.3 New money

The main new-money variable removes mechanical asset growth from returns and mergers. Under the end-of-quarter timing assumption,

$$\text{NewMoney}_{i,t} = \text{TNA}_{i,t} - \text{TNA}_{i,t-1}(1 + R_{i,t}) - \text{MGTNA}_{i,t},$$

where TNA is total net assets and MGTNA is the increase in TNA due to mergers. The paper also checks a beginning-of-quarter assumption and a percentage-flow definition; the major conclusions are unchanged.

5 Data

The paper uses a mutual fund database provided by Mark Carhart, Eugene Fama, and Kenneth French. The sample includes diversified U.S. equity funds from January 1970 through December

1993, the period over which quarterly total net asset values are available. The paper excludes sector funds, international funds, and balanced funds because their risk exposures differ from broad U.S. equity funds.

The sample contains 1,826 fund entities and 13,944 fund-years. There are on average 478 funds per month, with a minimum of 281 and a maximum of 1,196. The database includes defunct and merged funds, which is crucial because excluding disappearing funds would create survivorship bias.

The paper uses a "follow the money" approach for mergers: when a fund merges, investors are treated as moving into the surviving merger partner when that partner can be identified. This does not remove every possible source of bias, but it substantially reduces the usual survivorship problem.

6 Methodology

6.1 Grinblatt–Titman aggregate performance measure

The first main test uses the Grinblatt–Titman measure:

$$\mathrm{GT}_t = \sum_j R_{j,t+1}(w_{j,t} - w_{j,t-1}),$$

where $w_{j,t}$ is fund j's weight in the aggregate mutual fund portfolio at time t, and $R_{j,t+1}$ is the future return. If investors have no selection ability and expected returns are constant, changes in portfolio weights should be uncorrelated with future returns, so the measure should average to zero.

A positive GT measure means that investors increased weights in funds that subsequently performed well and reduced weights in funds that subsequently performed poorly. The GT measure is benchmark-free, but it is not directly implementable because it corresponds to a zero-cost portfolio that implicitly shorts some funds and buys others.

6.2 New money portfolios

The paper constructs eight portfolios. The most important are:

- Portfolio 3: equal-weighted funds with positive new money.
- Portfolio 4: equal-weighted funds with negative new money.
- Portfolio 5: positive new-money funds weighted by new money.
- Portfolio 6: negative new-money funds weighted by new money.
- Portfolio 7: equal-weighted funds above the median new-money flow.
- Portfolio 8: equal-weighted funds below the median new-money flow.

The difference between positive and negative portfolios is the key measure of smart money. The abnormal return of positive portfolios relative to the market is the key measure for the information effect.

6.3 Risk adjustment

The paper reports raw excess returns over the market, one-factor alphas, Fama–French three-factor alphas, and conditional versions of these alphas. The conditional measures use lagged public information variables such as the one-month Treasury-bill yield, dividend yield, bond default premium, and bond horizon premium. These conditional tests ask whether the flow effect is merely a response to public macro variables or time-varying style exposures.

7 Main empirical findings

7.1 Investors have relative fund-selection ability

The GT measure is positive and highly significant. This is the cleanest evidence that investors' buying and selling decisions are related to future relative fund performance. The effect remains significant after removing extreme observations.

7.2 Positive-flow funds outperform negative-flow funds

New-money portfolio tests confirm the GT result. Funds with positive cash flows outperform funds with negative cash flows over the subsequent quarter. The effect is especially robust in long-short comparisons such as Portfolio 3 minus Portfolio 4.

7.3 But positive-flow funds do not generally beat the market

For the whole sample, positive-flow portfolios do not generate statistically significant abnormal returns over the market. This is the distinction between smart money and information. Investors identify relatively better funds, but the aggregate long-only signal is generally not strong enough to deliver market-beating performance for all funds.

7.4 The effect is not just hot money chasing past winners

Investors do chase winners to some extent, and new-money portfolios are related to performance portfolios. But this relationship is not perfect. Figure 1 shows that the above-median new-money portfolio has a modest mean share of past winners of 53%, only slightly above a random 50% benchmark. Table V shows high correlation with performance strategies but also statistically meaningful differences.

7.5 Small funds drive the strongest results

The largest practical result is the size pattern. New-money signals are powerful for small funds and weak for large funds. Small-fund positive-flow portfolios earn materially higher alphas, and positive-minus-negative spreads are statistically significant. This suggests that investors' fund-specific information is most useful where information is less widely processed or where flows represent a larger signal.

7.6 Load and no-load funds differ

Load funds show a stronger smart money effect before load charges are deducted. This may reflect the higher trading cost: investors will only move into or out of load funds when their signal is strong enough. But because load charges reduce realized returns, no-load funds are more relevant for implementability. The paper finds some evidence that small no-load funds can produce abnormal returns using flow signals.

7.7 The signal decays quickly

The flow signal is short-lived. Holding-period results show that the positive-flow advantage declines over time, nearly disappears by around 24 to 30 months, and reverses after roughly 30 months. This is much shorter than long-horizon mean reversion and indicates that investors are forecasting short-term performance.

8 Economic interpretation

The results suggest that mutual fund investors collectively possess or aggregate fund-specific information. This information could come from better interpretation of fund performance, fund reputation, manager quality, marketing, fund-family information, or unobserved fund characteristics. The paper does not claim that every investor is sophisticated; rather, aggregate flows contain information.

The results also help rationalize continued investment in active mutual funds. Even if the average active fund does not beat the market, investors who shift money across funds can do better than the average fund. But the practical value is limited. The broad positive-flow strategy does not beat the market for the full sample, and the strongest implementable evidence is for small no-load funds.

9 Key limitations and caveats

- **The GT measure is not implementable.** It is useful for detecting selection ability, but it assumes zero-cost long-short reallocations across funds.
- **Short-sale constraints matter.** Mutual fund investors normally cannot short mutual funds, so long-short portfolio evidence is not directly tradable.
- **Flow data are delayed.** Public availability of cash flow information matters. The paper handles this with lagged strategies and finds weaker effects, except among small funds.
- **Load charges reduce realized returns.** Strong pre-load signals in load funds may not translate into actual after-load profits.
- **The sample ends in 1993.** The mutual fund industry, information environment, and competition among investors have changed since then.

10 Extracted figures and tables from the paper

10.1 Table I. GT Performance Estimates for the Aggregate Mutual Fund Portfolios: January 1970 to June 1993

Read. Table I reports the GT measure for aggregate mutual fund investors. The main estimate is 0.567% per month with a t-statistic of 4.57 and a Newey–West t-statistic of 3.41. After removing observations with absolute values greater than 5%, the estimate falls to 0.222% per month but remains highly significant.

Interpretation. This is the strongest evidence for the smart money effect. Aggregate investors increased portfolio weights in funds that later performed well and reduced weights in funds that later performed poorly. The outlier-adjusted result is important because it shows that the result is not only a few extreme months.

Caveat. The table does not show an implementable trading profit. The GT portfolio is a zero-cost object, not a feasible mutual fund allocation for most investors. It supports selection ability, not necessarily practical abnormal returns.

Table I

GT Performance Estimates for the Aggregate Mutual Fund Portfolios: January 1970 to June 1993

GT Performance measure = $(\Sigma_j R_{j,t+1}(w_{jt} - w_{j,t-1}))$, where $R_{j,t+1}$ is the return of fund j between time t and time $t-1$ and w_{jt} is the weight for fund j at time t. I calculate quarterly portfolio weights as the total net assets (TNA) of each fund divided by the sum of the TNA of all funds for each quarter from March 1970 to December 1993. The differenced weights (weights for quarter t minus weights for quarter $t-1$) are multiplied by the corresponding future monthly returns of the funds. For example, the July 1970 through September 1970 returns are multiplied by the difference between portfolio weights on June 30, 1970 and the weights on March 31, 1970. According to Grinblatt and Titman (1993), under the null hypothesis that investors have no selection ability, the GT measures are serially uncorrelated with mean zero. The t-statistic in column five is adjusted for autocorrelation using the Newey-West covariance matrix. The last column reports the probability that the absolute value of the Wilcoxon-Mann-Whitney Rank Z-statistic is greater than the observed Z-statistic under the null hypothesis. Performance measures are in percentage return per month.

	No. of Months	Performance Measure	t-Test Statistic	t-Test Newey-West	Wilcoxon Probability
All funds	282	0.567	4.57	3.41	0.00
Eliminating observations with absolute values greater than 5					
All funds	273	0.222	4.60	3.41	0.00

Table I. GT Performance Estimates for the Aggregate Mutual Fund Portfolios: January 1970 to June 1993

10.2 Table II. Performance of New Money Portfolios Estimated by Simple Excess Returns and Risk-Adjusted Returns Using the Portfolio Regression Approach: January 1970 to December 1993

Read. Table II compares positive-new-money and negative-new-money portfolios. The equal-weighted positive cash flow portfolio has an excess return of 0.025% per month, while the equal-weighted negative cash flow portfolio has -0.074% per month. The long-short difference, Portfolio 3 minus Portfolio 4, is 0.099% per month with a t-statistic of 2.93.

Interpretation. The table shows that investors move into future relative winners and out of future relative losers. The positive-flow funds do not significantly beat the market, but they do beat the average fund and significantly beat negative-flow funds. This is why the paper emphasizes relative selection ability rather than broad market-beating ability.

Economic point. The positive-minus-negative spread is not fully implementable because it requires shorting negative-flow mutual funds. However, it confirms that cash flow contains information about future fund performance.

Table II

Performance of New Money Portfolios Estimated by Simple Excess Returns and Risk-Adjusted Returns Using the Portfolio Regression Approach: January 1970 to December 1993

The excess return is calculated as $R_{it} - R_{mt}$, where R_{it} is the return of portfolio i between time t and time $t - 1$, and R_{mt} is market return, the return on a value-weighted portfolio of all stocks listed on the CRSP (Center for Research in Security Prices) database. The t-statistic in parentheses tests the performance difference between the particular portfolio and the average mutual fund. The t-statistics in brackets test the performance difference between the particular portfolio and the market. Alpha$_1$ is calculated from the time series regression of portfolio returns on the single factor model: $R_{pt} - R_{ft} = \alpha_p + \beta_p(R_{mt} - R_{ft}) + e_{pt}$. R_{pt} is the rate of return of portfolio p in month t, R_{ft} is the risk-free interest rate in month t, α_p is the excess return of the model, and β_p is the factor loading of the market factor. The t-statistics in parentheses test whether alpha$_1$ is significantly different from zero. Alpha$_3$ is calculated from the time-series regression of the excess portfolio returns on the excess market return and the mimicking returns for the size (SMB) and book-to-market equity (HML) factors: $R_{pt} - R_{ft} = \alpha_p + \beta_{p\mathrm{RMRF}}\mathrm{RMRF}_t + \beta_{p\mathrm{SMB}}\mathrm{SMB}_t + \beta_{p\mathrm{HML}}\mathrm{HML}_t + e_{pt}$. The excess market return, RMRF, is the difference between the return on a value-weighted CRSP stock portfolio and 1-month T-bill yields. SMB is the return on the mimicking portfolio for the common size factor in stock returns. HML is the return on the mimicking portfolio for the common book-to-market equity factor in stock returns. RMRF, SMB, and HML are constructed according to the descriptions in Fama and French (1993). β_p is the factor loading of the corresponding factor. EW denotes equally weighted, and CW means that the individual fund returns are weighted by their corresponding new money amount. New money = $\mathrm{TNA}_{it} - \mathrm{TNA}_{i,t-1} * (1 + \mathrm{R}_{it}) - \mathrm{MGTNA}_{it}$, where TNA is total net assets, and MGTNA is the increase in total net assets due to mergers. For Panel B, the t-statistics in parentheses test whether the performance difference between the positive and the negative portfolios is significantly different from zero, and the t-statistics in brackets are adjusted for autocorrelation using the Newey-West covariance matrix. Performance measures are in percentage return per month.

Portfolios	Excess Return	Alpha$_1$	Alpha$_3$
Panel A: Portfolio Performance			
1. Average fund	−0.046	−0.025	0.009
	(NA)	(−0.40)	(0.19)
	[−0.72]		
2. Weighted by total net asset	−0.058	−0.042	0.001
	(−0.27)	(−1.15)	(0.03)
	[−1.58]		
3. Positive cash flow (EW)	0.025	0.035	0.074
	(1.97)	(0.56)	(1.71)
	[0.41]		
4. Negative cash flow (EW)	−0.074	−0.053	−0.027
	(−0.76)	(−1.09)	(−0.70)
	[−1.49]		
5. Positive cash flow (CW)	0.003	0.002	0.089
	(0.85)	(0.03)	(1.54)
	[0.04]		
6. Negative cash flow (CW)	−0.103	−0.077	−0.039
	(−1.13)	(−1.73)	(−0.92)
	[−2.23]		
7. Upper 50 percent of all cash flow (EW)	−0.003	0.008	0.039
	(1.28)	(0.13)	(0.96)
	[−0.05]		
8. Lower 50 percent of all cash flow(EW)	−0.084	−0.060	−0.022
	(−0.96)	(−1.26)	(−0.55)
	[−1.71]		

Table II. New Money Portfolios, Part 1

Table II—*Continued*

Panel B: Performance Difference between the Positive and Negative Portfolios	Excess Return Mean Difference
Portfolio 3 − Portfolio 4	0.099
	(2.93)
	[2.66]
Portfolio 5 − Portfolio 6	0.106
	(1.62)
	[1.48]
Portfolio 7 − Portfolio 8	80.081
	(2.37)
	[2.21]

Table II. New Money Portfolios, Part 2

10.3 Table III. Performance of New Money Portfolios Estimated by Unconditional and Conditional Risk-Adjusted Returns Using the Fund Regression Approach: January 1970 to December 1993

Read. Table III repeats the analysis using fund-level risk adjustment and both unconditional and conditional alphas. Positive-flow portfolios have higher alphas than negative-flow portfolios. The equal-weighted positive-minus-negative spread is significant across all four alpha measures: 0.102, 0.092, 0.108, and 0.082% per month.

Interpretation. Conditional risk adjustment does not remove the smart money effect. This matters because one possible explanation is that investors simply load into styles or macro states that subsequently perform well. The conditional tests imply that the effect is not just public macro information or common style rotation.

Methodological point. The fund regression approach uses individual fund alphas, which helps account for time variation in portfolio composition. This complements the portfolio regression approach in Table II.

Table III

Performance of New Money Portfolios Estimated by Unconditional and Conditional Risk-Adjusted Returns Using the Fund Regression Approach: January 1970 to December 1993

Alpha_1 is calculated as the weighted average of the realized alphas of the individual funds obtained from the time-series regression: $R_{it} - R_{ft} = \alpha_i + \beta_i(R_{mt} - R_{ft}) + e_{it}$. R_{it} is the rate of return of fund i in month t; R_{mt} is market return, the return on a value-weighted portfolio of all stocks listed on the CRSP (Center for Research in Security Prices) database; R_{ft} is the risk-free interest rate in month t; α_i is the excess return of the model, and β_i is the factor loading of the market factor. Alpha_3 is calculated as the weighted average of the realized alphas of the individual funds obtained from the time-series regression: $R_{it} - R_{ft} = \alpha_i + \beta_{i\text{RMRF}}\text{RMRF}_t + \beta_{i\text{SMB}}\text{SMB}_t + \beta_{i\text{HML}}\text{HML}_t + e_{it}$. The excess market return, RMRF, is the difference between the return on a value-weighted CRSP stock portfolio and 1-month T-bill yields. SMB is the return on the mimicking portfolio for the common size factor in stock returns. HML is the return on the mimicking portfolio for the common book-to-market equity factor in stock returns. RMRF, SMB, and HML are constructed according to the descriptions in Fama and French (1993). β_i is the factor loading of the corresponding factor. The conditional alpha_1 is calculated as the weighted average of the realized alphas of the individual funds obtained from the time-series regressions: $R_{it} - R_{ft} = \alpha_i + \delta_{1i}(R_{mt} - R_{ft}) + \delta'_{2i}[z_{t-1} * (R_{mt} - R_{ft})] + e_{it}$. The conditional alpha_3 is calculated as the weighted average of the realized alphas of the individual funds obtained from the time series regressions: $R_{it} - R_{ft} = \alpha_i + \delta_{1i\text{RMRF}}\text{RMRF}_t + \delta_{1i\text{SMB}}\text{SMB}_t + \delta_{1i\text{HML}}\text{HML}_t + \delta'_{2i\text{RMRF}}(z_{t-1} * \text{RMRF}_t) + \delta'_{2i\text{SMB}}(z_{t-1} * \text{SMB}_t) + \delta'_{2i\text{HML}}(z_{t-1} * \text{HML}_t) + e_{it}$, where z is a vector of lagged predetermined instruments, and δ's are the factor loadings of the conditional regression factors. The t-statistic in parentheses tests the performance difference between the particular portfolio and the average mutual fund. The t-statistic in brackets tests the performance difference between the particular portfolio and the market. EW denotes equally weighted, and CW means that the individual fund returns are weighted by their corresponding new money amount. New money = $\text{TNA}_{it} - \text{TNA}_{i,t-1} * (1 + R_{it}) - \text{MGTNA}_{it}$, where TNA is total net assets, and MGTNA is the increase in total net assets due to mergers. For Panel B, the t-statistics in parentheses test whether the performance difference between the positive and the negative portfolios is significantly different from zero and the t-statistics in brackets are adjusted for autocorrelation using the Newey-West covariance matrix. Performance measures are in percentage return per month.

Portfolios	Alpha_1	Alpha_3	Conditional Alpha_1	Conditional Alpha_3
Panel A: Portfolio Performance				
1. Average fund	−0.039	0.004	0.006	0.037
	(NA)	(NA)	(NA)	(NA)
	[−0.76]	[0.12]	[0.12]	[1.19]
2. Weighted by total net asset	−0.041	0.013	−0.023	0.039
	(−0.08)	(0.52)	(−1.01)	(0.11)
	[−1.33]	[0.30]	[−0.75]	[1.22]
3. Positive cash flow (EW)	0.047	0.072	0.083	0.088
	(4.31)	(4.57)	(4.01)	(3.73)
	[0.80]	[1.85]	[1.45]	[2.51]
4. Negative cash flow (EW)	−0.055	−0.020	−0.025	0.006
	(−1.27)	(−2.80)	(−2.40)	(−3.67)
	[−1.15]	[−0.57]	[−0.54]	[0.19]
5. Positive cash flow (CW)	0.016	0.059	0.027	0.079
	(1.32)	(1.70)	(0.55)	(1.48)
	[0.25]	[1.37]	[0.46]	[2.04]

Table III. New Money Portfolios, Part 1

Table III—*Continued*

Portfolios	Alpha_1	Alpha_3	Conditional Alpha_1	Conditional Alpha_3
Panel A: Portfolio Performance (*continued*)				
6. Negative cash flow (CW)	−0.087	−0.031	−0.072	−0.002
	(−1.55)	(−1.87)	(−2.54)	(−2.29)
	[−2.08]	[−0.90]	[−1.75]	[−0.07]
7. Upper 50 percent of all cash flow (EW)	0.015	0.041	0.057	0.064
	(3.63)	(3.55)	(3.42)	(2.63)
	[0.27]	[1.11]	[1.30]	[1.91]
8. Lower 50 percent of all cash flow (EW)	−0.065	−0.017	−0.038	0.008
	(−1.57)	(−2.01)	(−2.67)	(−2.84)
	[−1.40]	[−0.47]	[−0.86]	[0.26]
Panel B: Mean Difference between the Positive and Negative Portfolios				
Portfolio 3 − Portfolio 4	0.102	0.092	0.108	0.082
	(3.54)	(4.42)	(3.89)	(4.30)
	[3.29]	[4.04]	[3.54]	[3.82]
Portfolio 5 − Portfolio 6	0.103	0.090	0.099	0.081
	(1.95)	(2.40)	(1.97)	(2.39)
	[1.81]	[2.35]	[1.81]	[2.28]
Portfolio 7 − Portfolio 8	0.080	0.058	0.095	0.055
	(2.69)	(3.03)	(3.27)	(3.07)
	[2.56]	[2.87]	[3.07]	[2.89]

than 50 percent. The rank correlation between new money deciles and dec-

Table III. New Money Portfolios, Part 2

10.4 Figure 1. Percentage of Past Winners in the Above-Median New Money Portfolio

Read. Figure 1 plots the percentage of funds in the above-median new-money portfolio that were past winners. The mean is 53%, only modestly above 50%.

Interpretation. Investors do chase past winners, but the degree of winner chasing is not large enough to explain the whole smart money effect mechanically. If new money were simply hot money chasing last quarter's winners, the percentage would be far above 50%. The figure therefore supports the idea that flows contain additional information beyond past returns.

Connection to Table V. Figure 1 is the visual motivation for comparing new-money strategies to repeat-winner strategies. The paper later shows that performance explains an important part of the time-series variation, but the two signals are not identical.

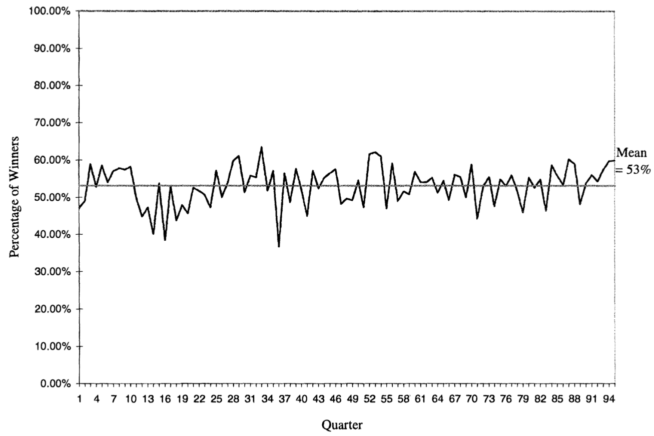

Figure 1. Percentage of past winners in the above-median new money portfolio. All funds are ranked by their returns and new money in the previous quarter. New money is defined as the dollar change in total net assets minus the appreciation in the fund assets and the increase in the total assets due to mergers. A fund is defined as a past winner if the return of the fund in the previous quarter is above the median performance of all funds in that same quarter. Otherwise, it is defined as a past loser. A fund belongs to the above-median new money portfolio if its amount of new money cash flow in the previous quarter is greater than the median cash flow of all funds in that same quarter. Otherwise, it belongs to the below-median new money portfolio. This figure plots the percentages of funds with above-median previous quarter returns in the above-median new money portfolio for each quarter from April 1970 to December 1993.

Figure 1. Percentage of Past Winners in the Above-Median New Money Portfolio

10.5 Table IV. Risk-Adjusted Returns of the Performance Portfolios: January 1970 to December 1993

Read. Table IV constructs portfolios based on past performance rather than new money. Positive-performance portfolios generally outperform negative-performance portfolios. Portfolio 3 minus Portfolio 4 has an excess return difference of 0.154% per month.

Interpretation. This table documents the repeat-winner effect: past winners tend to perform better than past losers over the short horizon. Since investors put more money into past winners, part of the smart money effect could simply reflect performance persistence.

Role in the paper. Table IV is not the final explanation. It is a benchmark against which the new-money strategy is compared. If new money only replicated past performance, then new money would add no incremental insight. Table V asks exactly how close the two strategies are.

Table IV

Risk-Adjusted Returns of the Performance Portfolios: January 1970 to December 1993

The excess return is calculated as $R_{it} - R_{mt}$, where R_{it} is the return of portfolio i between time t and time $t-1$, and R_{mt} is market return, the return on a value-weighted portfolio of all stocks listed on the CRSP (Center for Research in Security Prices) database. The t-statistic in parentheses tests the performance difference between the particular portfolio and the average mutual fund. The t-statistics in brackets test the performance difference between the particular portfolio and the market. Alpha_1 is calculated from the time series regression of portfolio returns on the single factor model: $R_{pt} - R_{ft} = \alpha_p + \beta_p(R_{mt} - R_{ft}) + e_{pt}$. R_{pt} is the rate of return of portfolio p in month t; R_{ft} is the risk-free interest rate in month t; α_p is the excess return of the model, and β_p is the factor loading of the market factor. The t-statistics in parentheses test whether alpha_1 is significantly different from zero. Alpha_3 is calculated from the time-series regression of the excess portfolio returns on the excess market return and the mimicking returns for the size (SMB) and book-to-market equity (HML) factors: $R_{pt} - R_{ft} = \alpha_p + \beta_{p\text{RMRF}}\text{RMRF}_t + \beta_{p\text{SMB}}\text{SMB}_t + \beta_{p\text{HML}}\text{HML}_t + e_{pt}$. The excess market return, RMRF, is the difference between the return on a value-weighted CRSP stock portfolio and 1-month T-bill yields. SMB is the return on the mimicking portfolio for the common size factor in stock returns. HML is the return on the mimicking portfolio for the common book-to-market equity factor in stock returns. RMRF, SMB, and HML are constructed according to the descriptions in Fama and French (1993). β_p is the factor loading of the corresponding factor. EW denotes equally weighted, and CW means that the individual fund returns are weighted by their corresponding new money amount. New money $= \text{TNA}_{it} - \text{TNA}_{i,t-1} * (1 + R_{it}) - \text{MGTNA}_{it}$, where TNA is total net assets, and MGTNA is the increase in total net assets due to mergers. For Panel B, the t-statistics in parentheses test whether the performance difference between the positive and the negative portfolios is significantly different from zero, and the t-statistics in brackets are adjusted for autocorrelation using the Newey-West covariance matrix. Performance measures are in percentage return per month.

Portfolios	Excess Return	Alpha_1	Alpha_3
Panel A: Portfolio Performance			
1. Average fund	−0.046	−0.025	0.009
	(NA)	(−0.40)	(0.19)
	[−0.72]		
2. Weighted by total net asset	−0.058	−0.042	0.001
	(−0.27)	(−1.15)	(0.03)
	[−1.58]		
3. Positive excess return (EW)	0.014	0.030	0.051
	(1.06)	(0.45)	(0.82)
	[0.21]		
4. Negative excess return (EW)	−0.140	−0.109	−0.073
	(−1.75)	(−1.76)	(−1.34)
	[−2.21]		
5. Positive excess return (CW)	−0.018	0.003	0.035
	(0.44)	(0.06)	(0.69)
	[−0.36]		
6. Negative excess return (CW)	−0.136	−0.103	−0.057
	(−1.36)	(−1.89)	(−1.05)
	[−2.39]		
7. Upper 50 percent of all excess return (EW)	0.014	0.026	0.050
	(1.15)	(0.38)	(0.84)
	[0.21]		
8. Lower 50 percent of all excess return (EW)	−0.138	−0.113	−0.072
	(−1.74)	(−1.65)	(−1.26)
	[−2.00]		

Table IV. Performance Portfolios, Part 1

Table IV—*Continued*

Panel B: Performance Difference between the Positive and Negative Portfolios	Excess Return Mean Difference
Portfolio 3 − Portfolio 4	0.154
	(1.82)
Portfolio 5 − Portfolio 6	0.117
	(1.53)
Portfolio 7 − Portfolio 8	0.153
	(1.77)

e new money cash flow as weights do better than the trading strategies at use equal weights for all funds. This observation suggests that the size relative size of cash flows helps to predict future fund returns within the

Table IV. Performance Portfolios, Part 2

10.6 Table V. Regressions Examining the Relationship between the Risk-Adjusted Excess Returns of the New Money Portfolios and the Performance Portfolios

Read. Table V regresses the risk-adjusted returns of new-money portfolios on the corresponding performance portfolios. The estimated slopes are around 0.827 to 0.883 and the R^2 values range from 0.66 to 0.76. Intercepts are close to zero.

Interpretation. A large part of the smart money strategy covaries with a repeat-winner strategy. This means performance chasing explains an important component of flow-based predictability. But the slopes are statistically different from one, and the portfolios are not identical. The paper therefore concludes that smart money is related to, but not completely explained by, chasing past winners.

Economic point. The table narrows the mechanism. Investors appear to use information correlated with past performance, but their flows are not a simple mechanical rank-and-buy rule based only on lagged returns.

10.7 Table VI. Performance of the New Money Portfolios for Small Funds and Large Funds: January 1970 to December 1993

Read. Table VI splits the sample by fund size. The results are much stronger for small funds. For small funds, the new-money weighted positive cash flow portfolio has a three-factor alpha of 0.181% per month, and the Portfolio 5 minus Portfolio 6 alpha spread is 0.258% per month. For large funds, the corresponding spread is only 0.073% per month and is not statistically strong.

Interpretation. This is one of the most important tables in the paper. The smart money effect is not uniform across the fund universe. It is concentrated in small funds, where investor flows may be more informative and where fund-specific information may be less efficiently incorporated.

Practical implication. If there is any implementable information effect, it is more likely to be found among small funds than among large funds. Large-fund flows contain much less useful signal.

Table V

Regressions Examining the Relationship between the Risk-Adjusted Excess Returns of the New Money Portfolios and the Performance Portfolios

$$\alpha^3_{\text{newmoney},t} = \alpha + \beta\alpha^3_{\text{performance},t} + e_t$$

$\alpha^3_{\text{newmoney},t}$ and $\alpha^3_{\text{performance},t}$ are calculated as the weighted average of the realized alphas of the individual funds obtained from the time-series regression: $R_{it} - R_{ft} = \alpha_i + \beta_{i\text{RMRF}}\text{RMRF}_t + \beta_{i\text{SMB}}\text{SMB}_t + \beta_{i\text{HML}}\text{HML}_t + e_{it}$. $\alpha^3_{\text{newmoney},t}$ is the risk-adjusted excess return of a portfolio of funds constructed according to their new money signals in the previous quarter. $\alpha^3_{\text{performance},t}$ is the risk-adjusted excess return of a portfolio of funds constructed according to their performance in the previous quarter. The t-statistics are in parentheses. The t-statistic for α tests whether it is different from zero and the t-statistic for β tests whether it is different from one.

Portfolios	α	β	R^2
Equally weighted positive new money portfolio on equally weighted positive performance portfolio	−0.000 (−0.28)	0.883 (−4.59)	0.70
Equally weighted negative new money portfolio on equally weighted negative performance portfolio	0.000 (1.56)	0.827 (−5.83)	0.66
Equally weighted above-median new money portfolio on equally weighted above-median performance portfolio	−0.000 (−1.39)	0.834 (−8.58)	0.76
Equally weighted below-median new money portfolio on equally weighted below-median performance portfolio	0.000 (2.09)	0.858 (−5.50)	0.68

Table V. New Money versus Performance Portfolios

Table VI

Performance of the New Money Portfolios for Small Funds and Large Funds: January 1970 to December 1993

Small funds are funds for which the total net assets (TNA) are smaller than the median TNA. Large funds are funds for which the total net assets are greater than the median TNA. The excess return is calculated as $R_{it} - R_{mt}$, where R_{it} is the return of portfolio i between time t and time $t-1$, and R_{mt} is market return, the return on a value weighted portfolio of all stocks listed on the CRSP (Center for Research in Security Prices) database. Alpha_3 is calculated from the time-series regression of the excess portfolio returns on the excess market return and the mimicking returns for the size (SMB) and book-to-market equity (HML) factors: $R_{pt} - R_{ft} = \alpha_p + \beta_{p\text{RMRF}}\text{RMRF}_t + \beta_{p\text{SMB}}\text{SMB}_t + \beta_{p\text{HML}}\text{HML}_t + e_{pt}$. The t-statistics in parentheses test the performance difference between the particular portfolio and the average mutual fund. The t-statistics in brackets test the performance difference between the particular portfolio and the market. EW denotes equally weighted, and CW means that the individual fund returns are weighted by their corresponding new money amount. For Panel B, the t-statistics in parentheses test whether the performance difference between the positive and the negative portfolios is significantly different from zero. Performance measures are in percentage return per month.

	Small Funds		Large Funds	
Portfolios	Excess Return	Alpha_3	Excess Return	Alpha_3
Panel A: Portfolio Performance				
1. Average fund	−0.028 (NA) [−0.30]	−0.009 (NA) [−0.25]	−0.065 (NA) [−1.33]	0.007 (NA) [0.19]
2. Weighted by total net asset	−0.022 (0.14) [−0.55]	0.016 (3.49) [0.38]	−0.061 (0.16) [−1.68]	0.009 (0.34) [0.27]
3. Positive cashflow (EW)	0.077 (1.74) [1.07]	0.099 (5.62) [2.24]	−0.021 (1.51) [−0.35]	0.045 (2.07) [1.15]
4. Negative cashflow (EW)	−0.112 (−1.36) [−1.65]	−0.064 (−3.59) [−1.61]	−0.048 (0.99) [−1.05]	0.015 (0.94) [0.40]
5. Positive cashflow (CW)	0.202 (2.84) [2.32]	0.181 (4.82) [3.17]	−0.020 (0.92) [−0.30]	0.045 (1.16) [1.03]
6. Negative cashflow (CW)	−0.139 (−1.34) [−1.53]	−0.077 (−1.58) [−1.31]	−0.101 (−1.11) [−2.23]	−0.028 (−1.91) [−0.79]
7. Upper 50 percent of all cashflow (EW)	0.054 (1.41) [0.79]	0.070 (5.06) [1.69]	−0.048 (1.01) [−0.87]	0.029 (2.10) [0.76]
8. Lower 50 percent of all cashflow (EW)	−0.108 (−1.27) [−1.57]	−0.051 (−2.68) [−1.22]	−0.056 (0.43) [−1.23]	0.010 (0.31) [0.28]
Panel B: Mean Difference between the Positive and Negative Portfolios				
Portfolio 3 − Portfolio 4	0.189 (4.65)	0.163 (5.50)	0.027 (0.65)	0.031 (1.23)
Portfolio 5 − Portfolio 6	0.342 (4.55)	0.258 (4.51)	0.081 (1.19)	0.073 (1.86)
Portfolio 7 − Portfolio 8	0.163 (4.30)	0.121 (4.35)	0.008 (0.24)	0.019 (0.95)

Table VI. Small Funds and Large Funds

10.8 Table VII. Performance of the New Money Portfolios for Load and No-Load Funds: January 1970 to December 1993

Read. Table VII separates load and no-load funds. Load-fund positive-flow portfolios generally have stronger performance than no-load positive-flow portfolios before deducting load charges. For load funds, positive-minus-negative spreads are significant across the main portfolio pairs. For no-load funds, evidence is weaker but still present for some strategies.

Interpretation. The stronger pre-load signal in load funds may arise because investors face higher trading costs and therefore only trade when their information is strong. However, the gross performance evidence must be interpreted carefully because loads reduce investors' realized returns.

Practical implication. This table highlights an important distinction between signal strength and tradability. Load fund flows may reveal stronger information, but no-load funds are more relevant for feasible abnormal returns.

10.9 Table VIII. Performance of the Lagged New Money Portfolios for All Funds and Small Funds: January 1970 to December 1993

Read. Table VIII allows for information delays by forming portfolios using lagged new-money signals. For all funds, delaying by a month weakens but does not fully eliminate the positive-minus-negative spread. For small funds, the effect remains much stronger. The small-fund new-money weighted positive portfolio has a three-factor alpha of 0.149% per month with a one-month lag and 0.137% with a one-quarter lag.

Interpretation. The lagged tests are crucial because investors cannot instantly observe all fund flow data. The full-sample information effect is weak after delays, but small-fund flows retain meaningful predictive content. This is where the paper finds the clearest evidence that flow information can be used for abnormal returns.

Practical implication. The evidence is most relevant for an investor who observes flow data with some delay and focuses on small funds. The signal is less useful for broad all-fund allocations.

Table VII

Performance of the New Money Portfolios for Load and No-Load Funds: January 1970 to December 1993

The excess return is calculated as $R_{it} - R_{mt}$, where R_{it} is the return of portfolio i between time t and time $t-1$, and R_{mt} is market return, the return on a value weighted portfolio of all stocks listed on the CRSP (Center for Research in Security Prices) database. Alpha_3 is calculated from the time-series regression of the excess portfolio returns on the excess market return and the mimicking returns for the size (SMB) and book-to-market equity (HML) factors: $R_{pt} - R_{ft} = \alpha_p + \beta_{p\text{RMRF}}\text{RMRF}_t + \beta_{p\text{SMB}}\text{SMB}_t + \beta_{p\text{HML}}\text{HML}_t + e_{pt}$. The t-statistics in parentheses test the performance difference between the particular portfolio and the average mutual fund. The t-statistics in brackets test the performance difference between the particular portfolio and the market. EW denotes equally weighted, and CW means that the individual fund returns are weighted by their corresponding new money amount. For Panel B, the t-statistics in parentheses test whether the performance difference between the positive and the negative portfolios is significantly different from zero. Performance measures are in percentage return per month.

	Load Funds		No-Load Funds	
Portfolios	Excess Return	Alpha_3	Excess Return	Alpha_3
Panel A: Portfolio Performance				
1. Average fund	−0.050	0.010	−0.033	0.019
	(NA)	(NA)	(NA)	(NA)
	[−0.87]	[0.26]	[−0.42]	[0.54]
2. Weighted by total net asset	−0.050	0.021	−0.050	−0.004
	(−0.01)	(0.75)	(−0.81)	(−0.85)
	[−1.36]	[0.59]	[−1.36]	[−0.11]
3. Positive cash flow (EW)	0.038	0.091	0.016	0.058
	(3.05)	(4.89)	(0.90)	(1.66)
	[0.62]	[2.16]	[0.23]	[1.39]
4. Negative cash flow (EW)	−0.063	−0.006	−0.056	−0.010
	(−0.48)	(−1.76)	(−0.41)	(−2.05)
	[−1.29]	[−0.17]	[−0.98]	[−0.27]
5. Positive cash flow (CW)	0.058	0.109	−0.059	0.015
	(2.13)	(3.02)	(−0.32)	(−0.08)
	[0.87]	[2.34]	[−0.67]	[0.27]
6. Negative cash flow (CW)	−0.091	−0.019	−0.075	−0.027
	(−0.92)	(−1.29)	(−0.63)	(−1.45)
	[−1.97]	[−0.52]	[−1.27]	[−0.59]
7. Upper 50 percent of all cash flow (EW)	0.013	0.067	−0.011	0.028
	(2.48)	(5.26)	(0.45)	(0.53)
	[0.21]	[1.66]	[−0.16]	[0.74]
8. Lower 50 percent of all cash flow (EW)	−0.076	−0.011	−0.063	−0.001
	(−0.86)	(−1.83)	(−0.52)	(−1.22)
	[−1.58]	[−0.31]	[−1.12]	[−0.03]
Panel B: Mean Performance Difference between the Positive and Negative Portfolios				
Portfolio 3 − Portfolio 4	0.101	0.097	0.072	0.068
	(2.93)	(4.21)	(1.53)	(2.00)
Portfolio 5 − Portfolio 6	0.149	0.128	0.016	0.042
	(2.44)	(3.37)	(0.18)	(0.65)
Portfolio 7 − Portfolio 8	0.088	0.079	0.053	0.029
	(2.63)	(3.82)	(1.21)	(0.94)

Table VIII

Performance of the Lagged New Money Portfolios for All Funds and Small Funds: January 1970 to December 1993

Small funds are funds for which the total net assets are smaller than the median TNA. Large funds are funds for which the total net assets are greater than the median TNA. Alpha_3 is calculated from the time series regression of the excess portfolio returns on the excess market return and the mimicking returns for the size (SMB) and book-to-market equity (HML) factors: $R_{pt} - R_{ft} = \alpha_p + \beta_{p\text{RMRF}}\text{RMRF}_t + \beta_{p\text{SMB}}\text{SMB}_t + \beta_{p\text{HML}}\text{HML}_t + e_{pt}$. The t-statistics in parentheses test the performance difference between the particular portfolio and the average mutual fund. The t-statistics in brackets test the performance difference between the particular portfolio and the market. EW denotes equally weighted, and CW means that the individual fund returns are weighted by their corresponding new money amount. For Panel B, the t-statistics in parentheses test whether the performance difference between the positive and the negative portfolios is significantly different from zero. Performance measures are in percentage return per month.

	Alpha_3			
	All Funds		Small Funds	
Portfolios	Lagged for a Month	Lagged for a Quarter	Lagged for a Month	Lagged for a Quarter
Panel A: Portfolio Performance				
1. Average fund	0.008	0.015	−0.007	0.000
	(NA)	(NA)	(NA)	(NA)
	[0.23]	[0.44]	[−0.20]	[0.01]
2. Weighted by total net asset	0.013	0.009	0.019	0.016
	(0.41)	(0.03)	(3.04)	(3.28)
	[0.41]	[0.28]	[0.46]	[0.38]
3. Positive cash flow (EW)	0.063	0.047	0.079	0.056
	(3.52)	(1.95)	(4.39)	(2.66)
	[1.60]	[1.27]	[1.77]	[1.32]
4. Negative cash flow (EW)	−0.007	0.005	−0.034	−0.000
	(−1.62)	(−1.18)	(−1.68)	(−0.05)
	[−0.20]	[0.14]	[−0.86]	[−0.01]
5. Positive cash flow (CW)	0.044	0.011	0.149	0.137
	(1.10)	(−0.13)	(4.32)	(4.07)
	[1.00]	[0.26]	[2.70]	[2.51]
6. Negative cash flow (CW)	−0.006	0.012	−0.013	0.039
	(−0.73)	(−0.20)	(−0.13)	(1.11)
	[−0.17]	[0.32]	[−0.23]	[0.73]
7. Upper 50 percent of all cash flow (EW)	0.033	0.037	0.059	0.048
	(2.27)	(1.99)	(4.00)	(2.78)
	[0.88]	[1.00]	[1.42]	[1.19]
8. Lower 50 percent of all cash flow (EW)	0.002	0.007	−0.026	0.001
	(−0.60)	(−0.79)	(−1.15)	(0.04)
	[0.05]	[0.20]	[−0.63]	[0.02]
Panel B: Mean Performance Difference between the Positive and Negative Portfolios				
Portfolio 3 − Portfolio 4	0.069	0.042	0.113	0.056
	(3.23)	(1.95)	(3.81)	(1.84)
Portfolio 5 − Portfolio 6	0.050	−0.000	0.162	0.098
	(1.31)	(−0.01)	(3.06)	(2.01)
Portfolio 7 − Portfolio 8	0.031	0.030	0.086	0.047
	(1.61)	(1.58)	(2.98)	(1.61)

Table VII. Load and No-Load Funds

Table VIII. Lagged New Money Portfolios

10.10 Table IX. Performance of the New Money Portfolios for Small Load and No-Load Funds: January 1970 to December 1993

Read. Table IX focuses on small funds and separates load from no-load funds. Small no-load funds show some of the strongest implementable results. For example, without lags, the small no-load new-money weighted positive cash flow portfolio has a three-factor alpha of 0.270% per month; with a one-month lag, the alpha remains 0.184% per month.

Interpretation. This table refines the practical conclusion. Small load funds display strong gross signals, but transaction costs are a barrier. Small no-load funds avoid load charges and still show positive abnormal returns, so they provide the best evidence for an implementable information effect.

Takeaway. The phrase “money is smart” is most accurate for small-fund flows, and especially for small no-load funds where the strategy is more feasible.

10.11 Figure 2. Performance of Positive and Negative Portfolios for Different Holding Periods

Read. Figure 2 plots the monthly average three-factor alpha of Portfolio 5 and Portfolio 6 as the holding period extends from 3 to 36 months. The positive-flow portfolio starts above the negative-flow portfolio, but the gap narrows steadily. Around 30 months the two portfolios are similar, and after that the negative portfolio performs better.

Interpretation. The smart money effect is short-run. Flows forecast near-term relative performance, not permanent manager superiority. This supports the idea that investors detect short-lived fund-specific information, performance persistence, or temporary opportunities.

Implication. A flow-based strategy must be rebalanced frequently. Long holding periods dilute and eventually reverse the signal.

Table IX

Performance of the New Money Portfolios for Small Load and No-Load Funds: January 1970 to December 1993

Small funds are funds for which the total net assets are smaller than the median TNA. Alpha_3 is calculated from the time-series regression of the excess portfolio returns on the excess market return and the mimicking returns for the size (SMB) and book-to-market equity (HML) factors: $R_{pt} - R_{ft} = \alpha_p + \beta_{p\text{RMRF}}\text{RMRF}_t + \beta_{p\text{SMB}}\text{SMB}_t + \beta_{p\text{HML}}\text{HML}_t + e_{pt}$. The t-statistics in parentheses test the performance difference between the particular portfolio and the average mutual fund. The t-statistics in brackets test the performance difference between the particular portfolio and the market. EW denotes equally weighted, and CW means that the individual fund returns are weighted by their corresponding new money amount. For Panel B, the t-statistics in parentheses test whether the performance difference between the positive and the negative portfolios is significantly different from zero. Performance measures are in percent return per month.

	Alpha_3			
	Small Load Funds		Small No-Load Funds	
Portfolios	Without Lags	Lagged for a Month	Without Lags	Lagged for a Month
Panel A: Portfolio Performance				
1. Average fund	−0.008	−0.007	0.027	0.025
	(NA)	(NA)	(NA)	(NA)
	[−0.20]	[−0.17]	[0.70]	[0.63]
2. Weighted by total net asset	0.007	0.005	0.042	0.050
	(1.62)	(0.99)	(2.16)	(1.89)
	[0.16]	[0.12]	[0.96]	[1.13]
3. Positive cash flow (EW)	0.103	0.087	0.122	0.108
	(4.52)	(4.09)	(2.99)	(2.05)
	[2.06]	[1.81]	[2.39]	[1.82]
4. Negative cash flow (EW)	−0.042	0.001	−0.062	−0.047
	(−1.72)	(0.40)	(−3.37)	(−2.68)
	[−0.95]	[0.03]	[−1.29]	[−0.96]
5. Positive cash flow (CW)	0.085	0.079	0.270	0.184
	(2.30)	(2.19)	(4.51)	(2.78)
	[1.41]	[1.38]	[3.80]	[2.42]
6. Negative cash flow (CW)	−0.045	0.009	−0.036	0.005
	(−0.70)	(0.32)	(−1.02)	(−0.32)
	[−0.68]	[0.14]	[−0.46]	[0.07]
7. Upper 50 percent of all cash flow (EW)	0.099	0.081	0.070	0.052
	(4.95)	(4.24)	(1.69)	(0.96)
	[2.11]	[1.76]	[1.54]	[1.08]
8. Lower 50 percent of all cash flow (EW)	−0.038	0.000	−0.025	−0.009
	(−1.42)	(0.36)	(−1.90)	(−1.16)
	[−0.80]	[0.01]	[−0.51]	[−0.18]
Panel B: Mean Performance Difference between the Positive and Negative Portfolios				
Portfolio 3 − Portfolio 4	0.145	0.086	0.185	0.155
	(3.73)	(2.37)	(3.79)	(2.74)
Portfolio 5 − Portfolio 6	0.130	0.070	0.306	0.179
	(1.83)	(1.05)	(3.70)	(2.16)
Portfolio 7 − Portfolio 8	0.137	0.081	0.095	0.062
	(3.55)	(2.31)	(2.03)	(1.21)

Table IX. Small Load and No-Load Funds

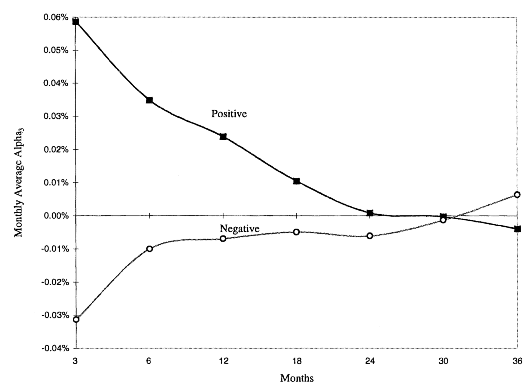

Figure 2. Performance of positive and negative portfolios for different holding periods. This figure plots the monthly average alpha_3 for portfolios 5 and 6 for different holding periods up to 36 months. Portfolio 5 invests in all available funds with positive new money cash flow and weights by funds' new money. Portfolio 6 invests in all available funds with negative new money cash flow and weights by funds' new money. Alpha_3 is calculated from the time-series regression of the excess portfolio returns on the excess market return and the mimicking returns for the size (SMB) and book-to-market equity (HML) factors. Then the average monthly portfolio alpha_3 is calculated for different holding periods of 1 to 36 months after the portfolio formation.

Figure 2. Performance for Different Holding Periods

11 Bottom line

Zheng (1999) finds that mutual fund investors display real but limited fund selection ability. New money flows predict short-term future relative performance, especially among small funds. The effect survives standard risk adjustment and conditional controls, and it is not merely a mechanical consequence of chasing past winners. However, the broad positive-flow strategy does not reliably beat the market in the full sample, and the strongest implementable evidence is concentrated in small no-load funds. The paper's most useful lesson is that fund flows are informative, but the information is short-lived, fund-size dependent, and not a general guarantee of market-beating returns.